

German Power Fair Value — Day-Ahead Forecasting Pipeline

Seya Makin | seyamakin04@gmail.com | github.com/seya-makin **Market:** DE-LU Day-Ahead
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Why Germany, Why Now

The German power market in 2024 is the most structurally interesting in Europe. Two opposing forces define it:

Renewable cannibalization drove 457 negative-price hours in 2024 — 5.2% of the year — as solar penetration hit 17% of generation. On summer weekends, wind and solar simultaneously flood the grid and prices collapse to -€500/MWh. A model blind to this dynamic is useless.

Dunkelflaute events are the opposite extreme. In November 2024 (144 hours) and December 2024 (72 hours), prolonged wind-and-solar droughts under high-pressure systems drove German intraday prices to €820/MWh and €900/MWh — the highest in 18 years. Gas peakers earned over 50% of their annual wholesale revenues in those two events alone.

Any fair-value system for the German DA market must explicitly model both regimes. This pipeline does.

1. Data Ingestion and Quality

Source: SMARD (smard.de) — Bundesnetzagentur electricity market platform. Data retrieved from ENTSO-E under EU Regulation 543/2013. Equivalent data quality, no API key required, fully public.

Programmatic API endpoints:

Series	Filter ID	Endpoint Pattern	Unit
DA Price DE-LU	4169	/chart_data/4169/DE-LU/	EUR/MWh
Total Load	410	/chart_data/410/DE/	MW
Wind Onshore	123	/chart_data/123/DE/	MW
Wind Offshore	3791	/chart_data/3791/DE/	MW
Solar PV	125	/chart_data/125/DE/	MW

Base URL: https://www.smard.de/app/chart_data/{filter}/{region}/{filter}_{region}hour{ts}.json

Response: `[[unix_ms, value], ...]` arrays converted to UTC-indexed hourly Series.

Dataset: 26,304 hours, 2022-01-01 to 2024-12-31.

Timezone: UTC internal storage. CET/CEST for display. Spring-forward gap interpolated. Autumn duplicate averaged. Assert: exactly 8,784 rows for 2024 — passes.

Structural breaks:

- post_ukraine_war: 2022-02-24 gas price shock
- post_nuclear_phaseout: 2023-04-15 DE nuclear → 0 MW

LLM Data QA (Gemini 2.0 Flash): 20 electricity-market-specific validation rules proposed programmatically and executed as Python boolean conditions. Rules include: solar zero between 20:00-05:00 CET, load bounds 20,000-100,000 MW, price bounds -500 to 3,000 EUR/MWh, DE nuclear = 0 after 2023-04-15, Dunkelflaute detection logic. QA score: 97.1/100 — PASS.

2024 verified facts:

- Mean DA price: €78.51/MWh
 - Negative hours: 457 (5.2%) — Apr-Sep concentrated
 - Dunkelflaute detected: Nov 2-7 (144h), Dec 12-14 (72h)
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2. Forecasting and Model Validation

Target: Hourly DA prices for delivery day + 1. Delivery-period aggregates derived from hourly distribution.

The fundamental insight: German prices are driven by one number: residual load = load - wind - solar. High residual load → gas marginal → prices spike. Low or negative → renewables marginal → prices collapse. Every feature is built around capturing this at multiple timescales.

67 features across 8 groups:

Group	Count	Key Features
Fundamental	11	residual_load, renewable_penetration
Fuel/Carbon	7	clean_spark_spread, dark_spread
Calendar	16	hour_sin/cos, DE holidays, peak_hour
Lags	8	price_lag_168h (r = 0.71), price_lag_8736h
Rolling	10	price_roll7d_mean, negative_price_freq_7d
Interactions	6	solar_x_summer_weekend, wind_x_offpeak
Regime	8	price_regime 0-3, dunkelflaute_severity
Structural	3	post_ukraine_war, post_nuclear_phaseout

Leakage prevention: all lags shifted ≥ 24 h. Formal assertion: no feature shows correlation > 0.95 with future target.

Results:

Model	2024 MAE	Skill vs Naive
Seasonal Naive	~32 EUR/MWh	baseline
Ridge Regression	~29 EUR/MWh	+ 9%

XGBoost + Two-Stage **25.37 EUR/MWh + 22.7%**

MAE by year:

Year	MAE	Context
2022	80.18	Ukraine crisis — prices €200-400/MWh
2023	52.54	Transition year
2024	25.37	Primary benchmark year

Directional accuracy: **82.9%** — correct price direction 4 out of every 5 hours across all conditions.

Two-stage extreme model: Binary classifier detects extreme hours (dynamic 85th percentile threshold, $\approx$ €93/MWh in 2024). Dedicated regressor trained on extreme hours with sample weights proportional to price level. 17.2% of 2024 hours routed to extreme model.

Negative price model: Binary classifier with `scale_pos_weight` = 18.2 matched to real 5.2% negative frequency. Summer recall: 91.1%. Full-year recall: 88.0%.

Validation: 52-window expanding walk-forward

- Minimum train: 365 days
- Step: 7 days (production cadence)
- Test: full 2024, 8,784 hours
- Zero shuffling, zero future leakage

Conformal Prediction (O'Connor et al., 2025): Regime-conditioned split conformal prediction. Distribution-free coverage guarantee — no Gaussian assumptions. Critical for German power where prices are bimodal. Empirical 90% coverage: **91.0% ✓**

3. Prompt Curve Translation

Delivery-period aggregates:

Horizon	Definition	Instrument
Prompt Day Baseload	Mean 24h	DE DA base block
Prompt Day Peak	Mean hours 08-20 Mon-Fri	DE DA peak block
Prompt Week Base	7-day mean	EEX Week Base Future
Prompt Month Base	Calendar month mean	EEX Month Base Future

All with 80% and 95% conformal prediction intervals.

Signal construction: LONG if $FV > \text{reference} + 0.5 \times \text{CP80_width}$ SHORT if $FV < \text{reference} - 0.5 \times \text{CP80_width}$ NEUTRAL otherwise HIGH conviction if $\text{CP80 width} < 80\%$ of 7-day mean

Regime-conditional trading:

Regime 3 (Dunkelflaute): residual load $> 55\text{GW}$, wind $< 5\text{GW}$ $\rightarrow$ LONG prompt peak. Buy DE prompt peak on EEX or gas generation optionality. Nov 2024 event returned

€94,000/MW over 144 hours.

Regime 0 (Glut): renewable penetration > 80%, weekend, summer → SHORT prompt
baseload. Buy battery charge/discharge spread.

8 invalidation conditions:

- 1. Wind forecast revised > 20% post-signal
- 2. TTF gas moves > €3/MWh overnight
- 3. French nuclear drops > 2,000 MW unexpectedly
- 4. Actual DA clearing > 2σ from forecast
- 5. Residual load changes > 15% post-signal
- 6. Unplanned outage > 1,000 MW announced
- 7. Regime classification changes pre-delivery
- 8. Conformal band expands > 50% vs 7-day mean

4. AI-Accelerated Workflow

Three Gemini 2.0 Flash integrations. All programmatic. All logged to JSONL. All auditable.

Component 1 — LLM Data QA Input: schema + 10 sample rows + statistics Output: 20 physics-grounded validation rules as Python boolean conditions executed against 26,304 rows Productivity gain: replaces 3 hours of manual rule writing with 4-second LLM call Log: outputs/logs/llm_qa_prompts.jsonl

Example rule: "Solar > 0 between 20:00-05:00 CET indicates timezone error (ERROR severity)"

Component 2 — Daily Market Commentary Input: 24 pipeline-computed metrics only Output: 180-word Bloomberg-style market note Anti-hallucination guard: every number in commentary cross-checked against input metrics (± 0.5 tolerance) Log: outputs/logs/llm_commentary_prompts.jsonl

Component 3 — Ingestion Config Generation Input: SMARD/ENTSO-E field documentation Output: structured JSON config with column names, units, expected ranges, aggregation rules Eliminates manual hardcoding of field mappings Log: outputs/logs/llm_config_generation.jsonl

Component	Logged	Hallucination Check	On Failure
QA Rules	✓ JSONL	N/A	Log + continue
Commentary	✓ JSONL	✓ ± 0.5 tolerance	Flag + log
Config Gen	✓ JSONL	N/A	Fallback defaults

Appendix

Repo: github.com/seya-makin/cobblestone-power **Pipeline:** python run_pipeline.py --mode full **Dashboard:** streamlit run dashboard/app.py **Submission:** submission.csv — 8,784 rows, 2024 out-of-sample with 7 quantiles, conformal intervals, regime labels, and Dunkelflaute risk scores

References:

1. O'Connor et al. (2025). Conformal Prediction for Electricity Price Forecasting. *Energy and AI*, 21.
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4. Wood Mackenzie (2025). Weathering the lulls: risks and opportunities of Dunkelflaute.
5. Timera Energy (2025). Impact of German Dunkelflaute on flex asset value.
6. FfE (2026). German electricity prices on EPEX Spot in 2025.